SoXR No-Code XR Toolkit Guide

By Sarah Gerhart

What is the Toolkit?

The Toolkit is a collection of four components designed to allow team members to create interactive and reactive scenes in unity without having to write any code. You can simply attach these general-purpose components to any GameObject and connect them in Unity's Inspector!

3 steps to adding any component:

1. Select any object in your scene, click "Add Component" at the bottom of the inspector, and select the desired component from the SoXR Toolkit menu (or just drag the script onto the object).
2. Configure the settings (like countdowns or tag names) in the inspector using the text fields and tooltips.
3. Look for the UnityEvents section of the component, click the + icon, drag in the target object, and pick what action should happen when the event fires.

XR Grab Interactable (the built-in one)

What it does

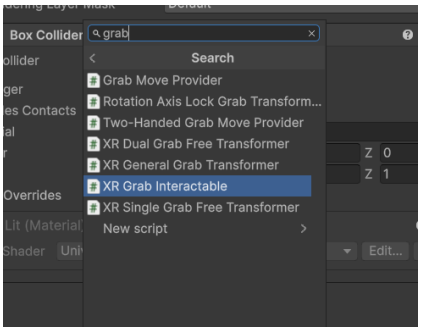
- Allows users to pick up, hold, throw, and interact with objects using their controllers or hands.

When to use it

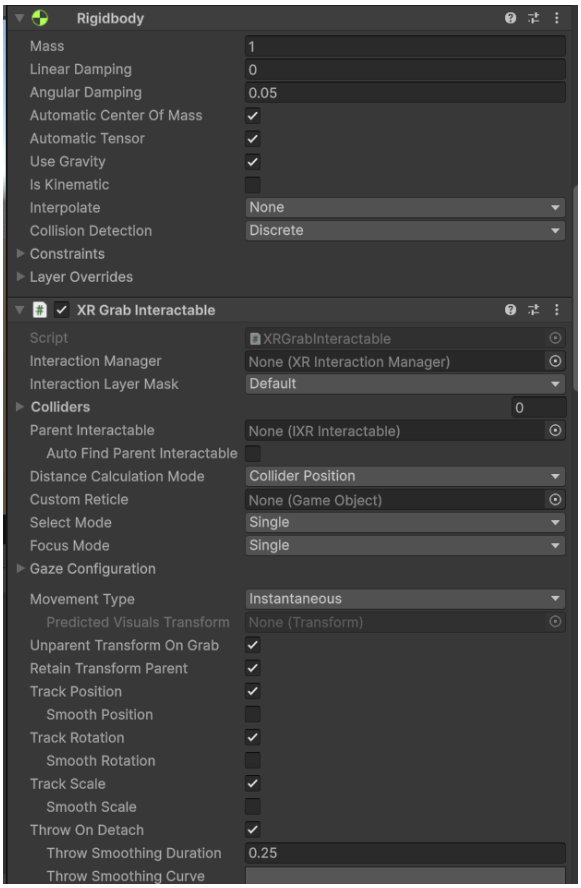
- Whenever an object needs to be pick-up-able, such as throwing a ball or grabbing a mug.

How to set it up

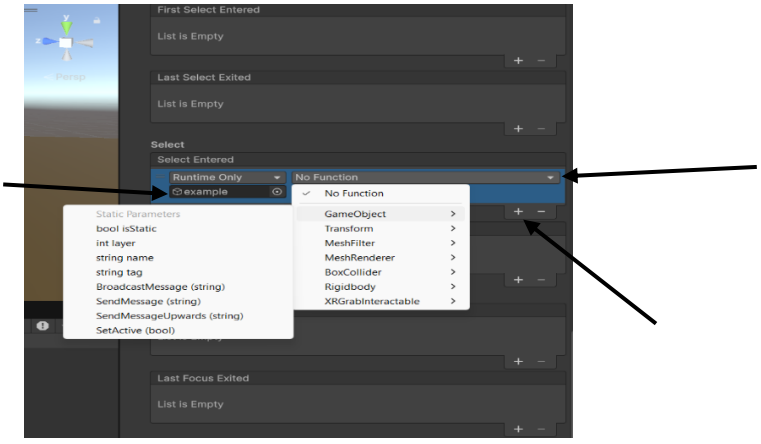
1. Click "Add Component" within your 3D object's inspect menu and search for "XR Grab Interactable"



2. It will automatically make a Ridgidbody along with adding the grab component



3. Now you can change any settings necessary (see below for descriptions on our most used setting options) and add an interactable event.
4. Scroll to the “Interactable Events” section and click the + icon on the event you want.
5. Drag the object into the designated object location (marked by example in the image below) and link the function and action you want to happen by clicking the option you want in the “Static Parameters” Window (in the image the GameObject function is being hovered over to show its options)



6. Now you have an object that can be picked up and cause something to happen!

Setting Details

Common Settings	**What it does**
Movement Type	*Velocity Tracking* (default) feels physical and the object pushes into walls; *Kinematic* follows your hand but still collides; *Instantaneous* snaps to your hand with no delay.
Throw On Detach	If checked, releasing the object while moving your hand throws it.
Attach Transform	Allows you to define where and at what angle the object sits in the user's hand. Great for tools and handles. Right click your object in the hierarchy and click create empty- give it a name like hold. Back in the main object's inspector, put hold into the object spot for attach transform and adjust its rotation until it sits as desired.
Select Mode	*Single* means one hand at a time; *Multiple* allows two-handed grabs and hand-offs.
Interaction Layer Mask	Controls which interactors (hands, rays, sockets) are allowed to grab this object.
Interactable Event: Hover Entered / Exited	Fires when a hand or ray starts/stops pointing at the object.
Interactable Event: Select Entered / Exited	Fires when the object is grabbed / released.
Interactable Event: Activated / Deactivated	Fires when the object is held and the player presses/releases the trigger button.

Common Mistakes

- If your object falls through the floor, the ground has no Collider. A Plane or Cube from the GameObject → 3D Object menu has one by default; an empty object or imported mesh may not.
- If you're adjusting the hold angle for attach transform, make sure you have the empty object linked into the grab objects settings.
- If you can't reach the object make sure you are close enough to the object and the ray from the controller is pointing at the object

On Trigger Action

What it does

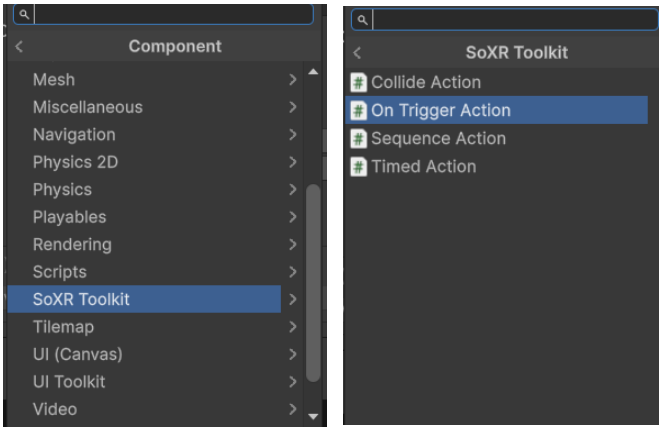
- Fires customizable events when a specific object (like the player's headset) enters or exits a designated invisible boundary area.

When to use it

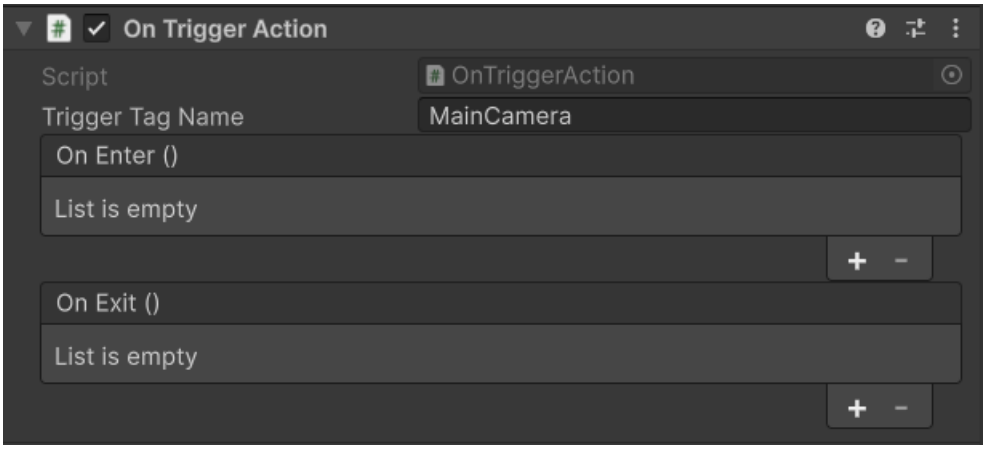
- Any time you want something to be triggered based on proximity. For example, having a welcome message pop up when a user enters a room.

How to set it up

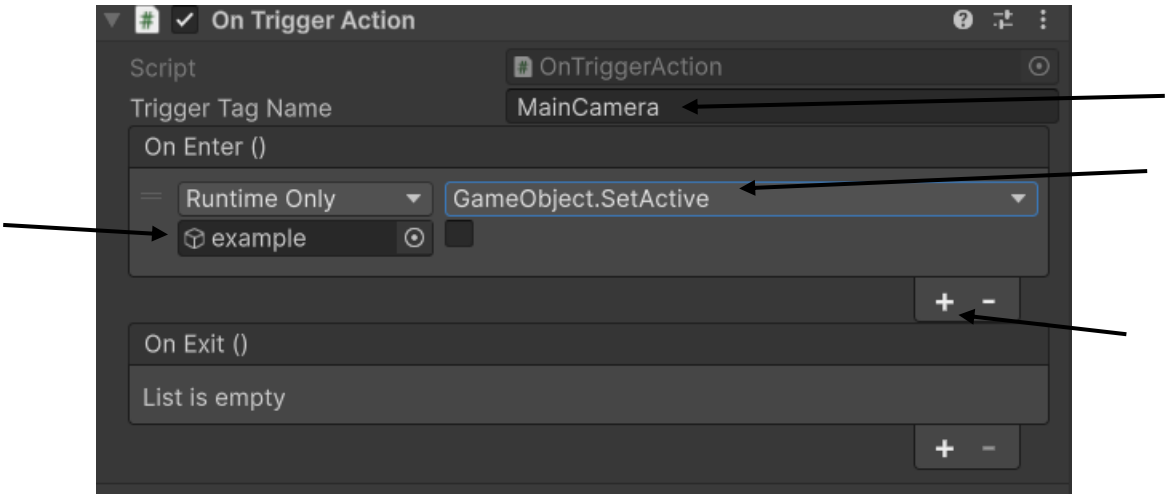
1. Click "Add Component" within your object's inspect menu, click the SoXR Toolkit menu, and click onTriggerAction. (The object can be invisible or a 3D object, for an invisible trigger location, right click in the hierarchy window and click "create empty," then add a box collider with isTrigger checked and shape and place it however you need.)



2. A menu as shown below will be added to your object and now you can add specific settings.



3. Set the trigger tag name to the object you want to cause the trigger action. It defaults to MainCamera which is the headset, so you will likely not need to change this as that is a good trigger.
4. Next, click the + icon on either the On Enter section or On Exit section depending on if you want an action to be triggered when entering or exiting the zone.
5. Click and drag the object you want to be affected by the trigger to the object designation and select the function you want to link. In the image below, example is the object being affected and it will disappear when the user runs into it.



6. Now you have a trigger for any action you want!

Setting Details

Common Settings	What it does
Trigger Tag Name	The tag of the trigger for the action. Recommended Default: MainCamera
On Enter ()	Fires when the trigger enters the specified zone.
On Exit ()	Fires when the trigger exits the specified zone.

Common Mistakes

- If the trigger doesn't fire, check that isTrigger is checked on the collider, that the entering object has the exact matching tag, and that at least one object has a rigidbody.
- Make sure the object you want to be affected by the trigger is the one linked in the settings.
- If using an invisible collider, make sure it is placed where you thought it was.

Timed Action

What it does

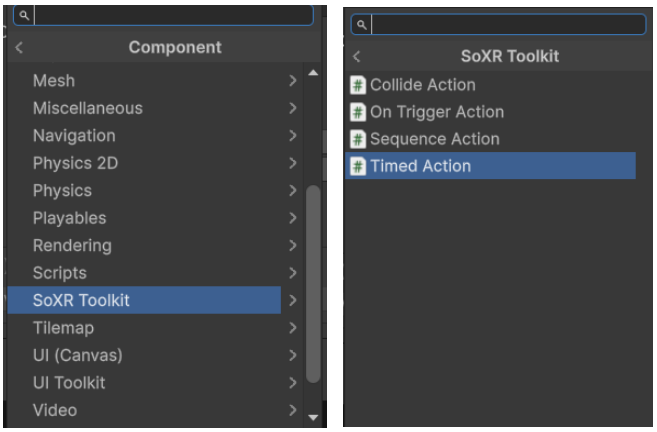
- Counts down a specific number of seconds and fires events once time runs out, with options to run once or repeat.

When to use it

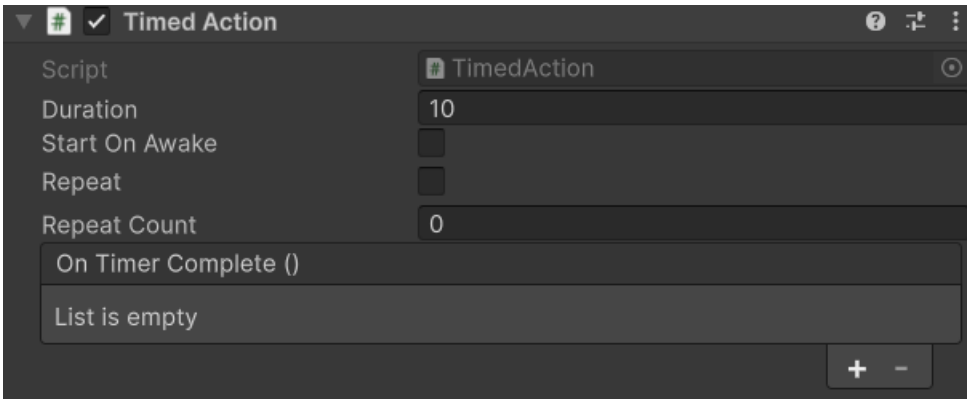
- Use when you want to trigger a timed event, such as revealing a hint if the user hasn't progressed in 30 seconds or spawning something after a certain amount of time.

How to set it up

1. Click "Add Component" within your object's inspect menu, click the SoXR Toolkit menu, and click Timed Action.

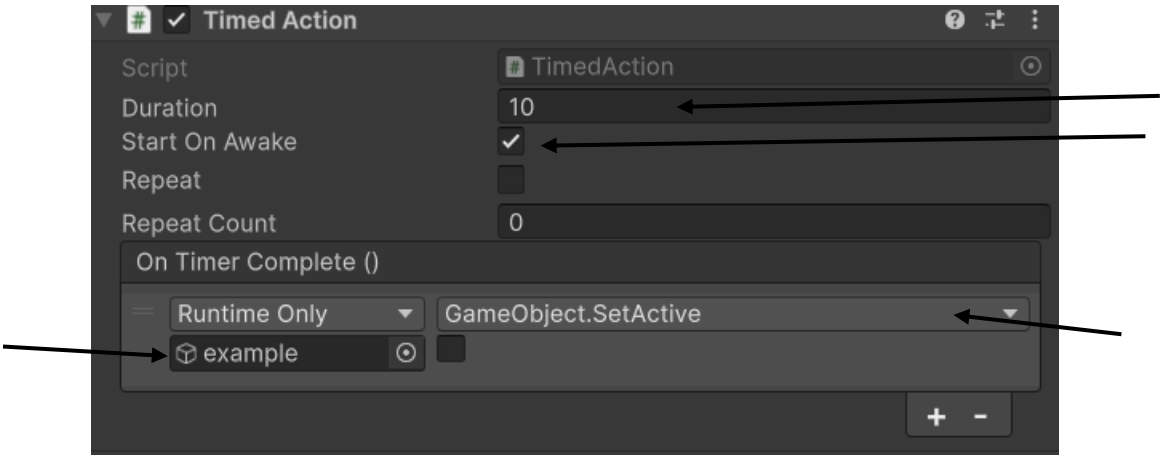


2. A menu as shown below will be added to your object and now you can add specific settings.



3. Set the duration to however long you want the action to be delayed

4. Select start on awake if you want your countdown to start as soon as the object is active. If this is not selected, the timed action will need to be triggered (maybe with one of the other toolkit actions)
5. Check repeat and the number of repetitions if you want the timer to run multiple times.
6. Next, click the + icon in on timer complete and add your object and action as described above.
7. The settings below will cause the example object to disappear 10 seconds after the scene starts.



8. Now you have a timer for any action you want!

Setting Details

Common Settings	**What it does**
Duration	The length of the countdown in seconds.
Start On Awake	If checked, starts the countdown immediately upon the scene load. Recommended Default: unchecked (unless auto starting)
Repeat	If checked, restarts the timer automatically after completion Recommended Default: unchecked
Repeat Count	Number of times to restart. Set to 0 to loop forever. Recommended Default: 0
On Timer Complete ()	Fires when the countdown finishes

Common Mistakes

- If the timer doesn't start double check if start on awake is unchecked. If it is, make sure another component is explicitly calling StartTimer().

Collide Action

What it does

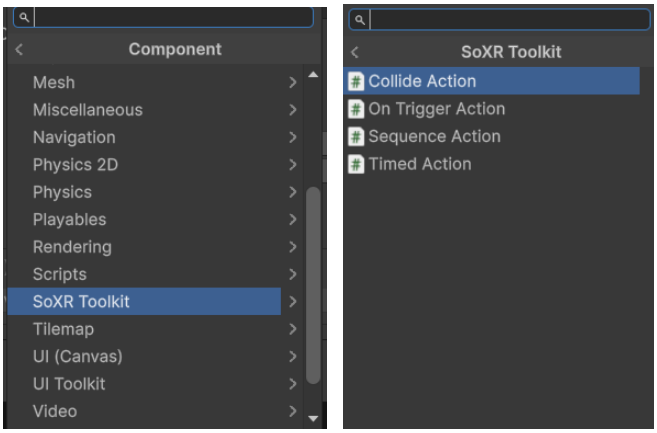
- Fires customizable events when a physical object crashes into another solid object.

When to use it

- Use when you want two solids to interact such as throwing an object at a target.

How to set it up

1. Click "Add Component" within your object's inspect menu, click the SoXR Toolkit menu, and click Collide Action.

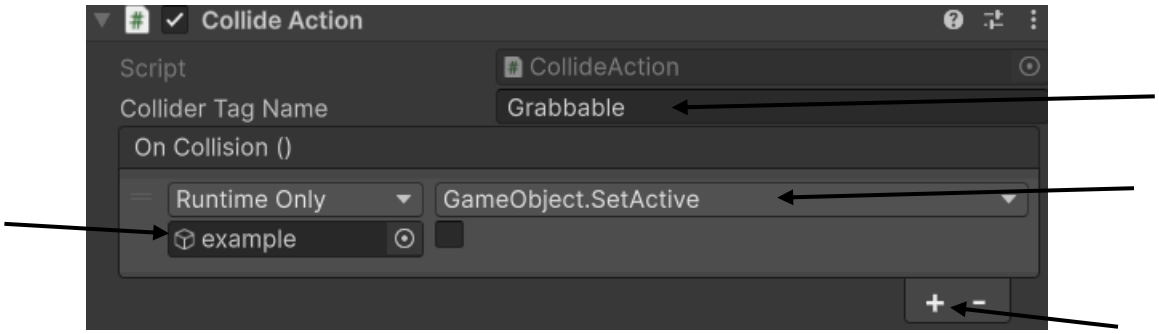


2. A menu as shown below will be added to your object and now you can add specific settings.



3. Set the collider tag name to the object that will be moved to cause the collision.

4. Next, click the + icon in on collision and add the other object part of the collision and the action as described above.
5. The settings below will cause the example object to disappear when the object with the tag "Grabbable" collides with it.



6. Now you have a collision action!

Setting Details

Common Settings	What it does
Collider Tag Name	The required tag of the object that must crash into this one to trigger the event. Recommended Default: Grabbable
On Collision ()	Fires upon physical contact

Common Mistakes

- If nothing happens on impact, make sure trigger is unchecked on both objects. If trigger is checked, objects pass through each other like ghosts, preventing physical collisions. Also, double check the tag matches to collider exactly and that at least one of the objects has a rigidbody.
- If there is no impact, make sure both objects have a collider in the first place.

Sequence Action

What it does

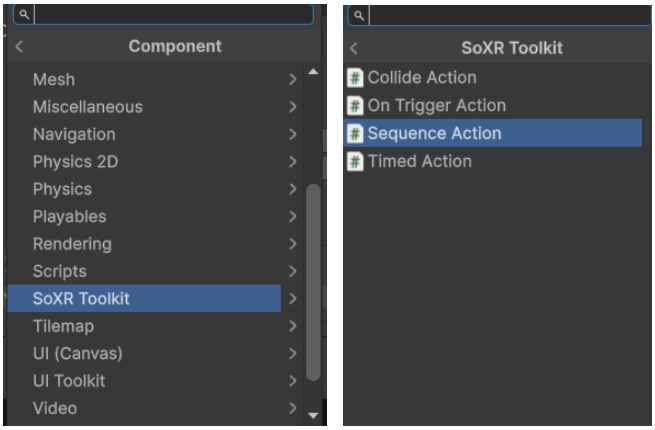
- Executes a list of multiple actions one after another in order, with custom time delays between each step.

When to use it

- Use when you want a multi-step series of events to occur. For example, the user walks through a door causing text to appear and then a rumbling sound plays after they have had time to read and the door is gone.

How to set it up

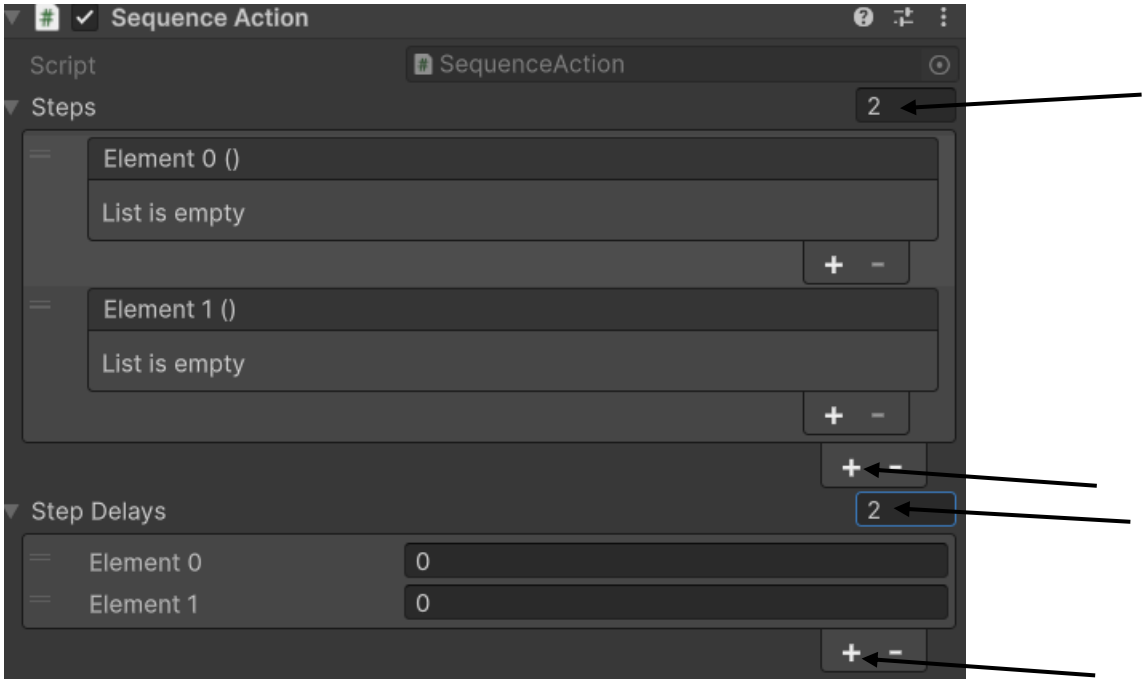
1. Click “Add Component” within your object’s inspect menu, click the SoXR Toolkit menu, and click Sequence Action.



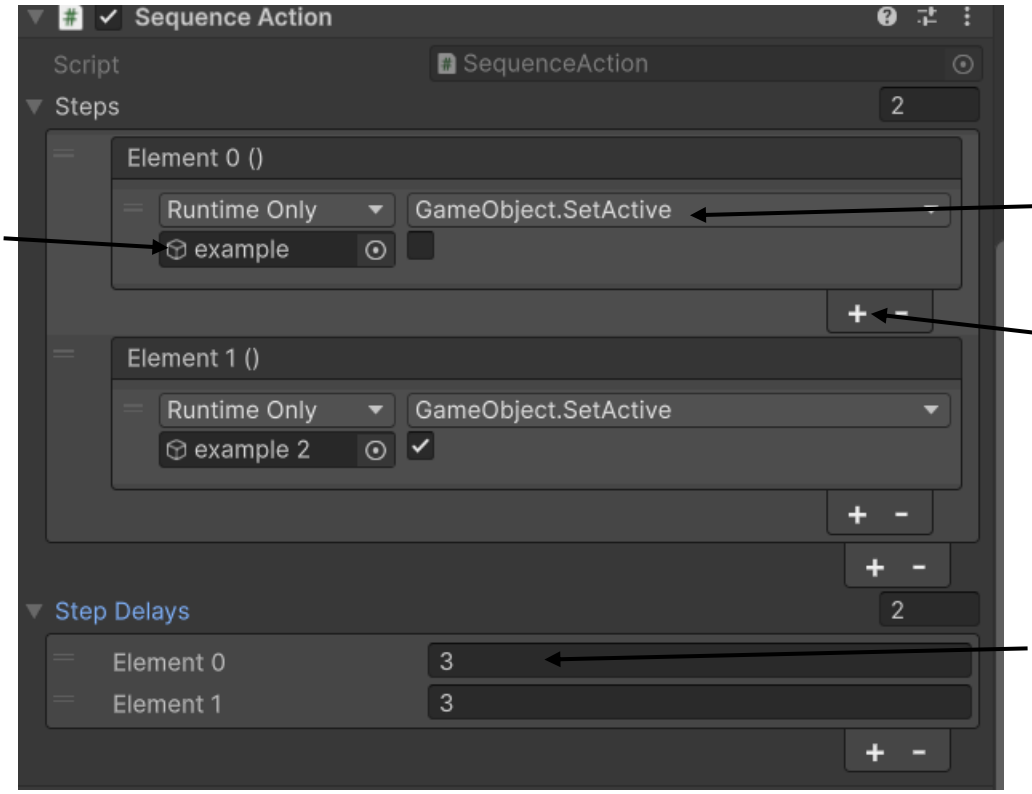
2. A menu as shown below will be added to your object and now you can add specific settings.



3. Set the number of steps you want in your sequence by either typing the number in the box or clicking the + icon.
4. Add the number of delays you want as well, if you have less than there are steps, the later steps will not have delays.
5. When adding 2 steps and delays the menu will change to look like the following.



6. Next, click the + icon in each element and add objects and actions as described above.
7. Add delays for the corresponding elements (element 0 in delays is how long until element 0 in steps fires)
8. The example below has the example object disappearing after 3 seconds and then example 2 appearing after 3 more seconds.



9. Now you have a sequence of actions!

Setting Details

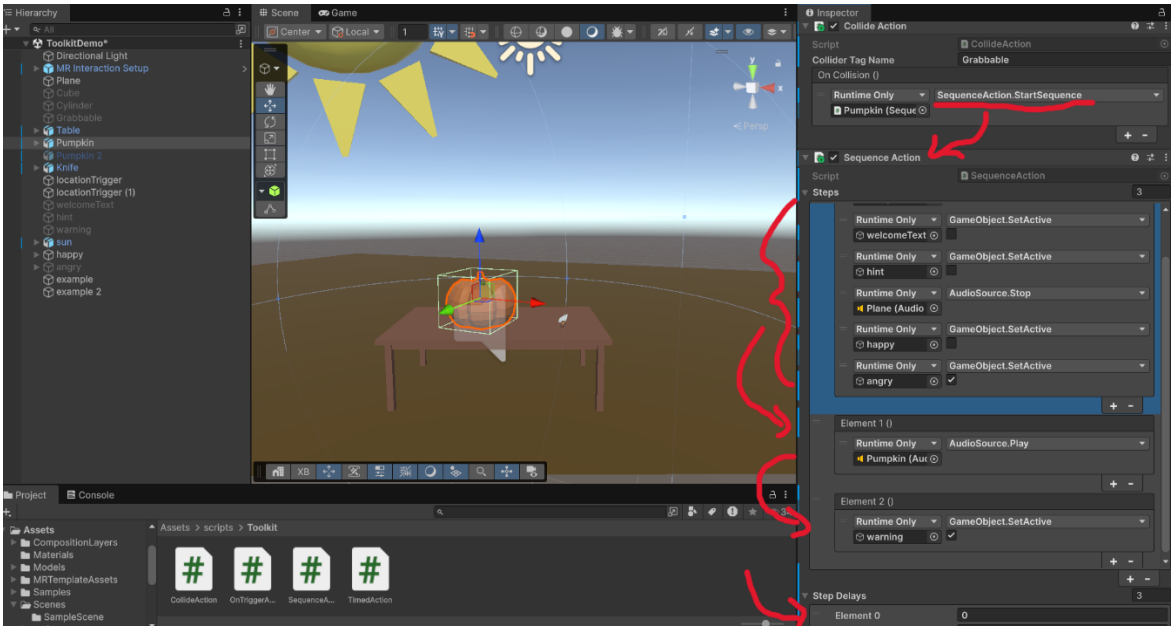
Common Settings	What it does
Steps	A list of UnityEvents to be executed in sequential order
Step Delays	A list of times in seconds representing how long to wait before each step fires. Recommended: Match the number of steps

Common Mistakes

- If the sequence doesn't play remember that the sequence waits for an external call to StartSequence(). Make sure a trigger or timer is hooked up to invoke SequenceAction.StartSequence().

Putting it Together

- I created an example scene that puts some of these components together.
- As shown below, I have a scene with a pumpkin and a knife on a table
- The Knife has the grabbable tag and has a collision action set up with the pumpkin that triggers the sequence action on collision.
- The sequence of events then happens in order with a 1 second delay between each (the delay is cut off in the image).



- The pumpkin can also be picked up, so, with the select exited event, after the pumpkin is dropped, a timed action begins that will respawn another pumpkin on the table and do a few other actions.

